

MATH 231 – Linear Algebra I

The Vector Unit, Part 3 ■ Cauchy–Schwarz, Metrics, and Norms

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About this document. This is Part 3 of the vector unit. Section numbering runs continuously through all three parts, so a reference such as §12.6 points into Part 2 and means exactly what it says:

Part	Sections	Subject
1	§1–§9	vectors, notation, addition, scaling, equality
2	§10–§12	orientation, the dot product, projection
3	§13–§15	Cauchy–Schwarz, metrics, norms
4	§16–	the standard basis, span, coordinates

Assumed from Parts 1–2. The Euclidean norm and the scaling rule $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$ (§8); the triangle inequality $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (§7.2); the dot product and $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$ (§11); the angle $\theta \in [0, \pi]$ between two vectors (§10.3); and the projection $\text{proj}_{\mathbf{u}}(\mathbf{v})$ with its component $\text{comp}_{\mathbf{u}}(\mathbf{v})$ (§12.4).

What you should be able to do afterwards. State the Cauchy–Schwarz inequality and prove it two ways; list the axioms a metric must satisfy and verify them for the Euclidean distance; compute Manhattan and Chebyshev distances; explain why the 1-, 2- and ∞ -norms carry those subscripts, and derive the ∞ case as a limit; state the three properties that define a norm; and state Hölder’s inequality and recognise Cauchy–Schwarz as its self-conjugate case.

Further reading. Boyd & Vandenberghe, *Introduction to Applied Linear Algebra*, Ch. 3 (norm and distance); Axler, *Linear Algebra Done Right*, Ch. 6 (Cauchy–Schwarz); Trefethen & Bau, *Numerical Linear Algebra*, Lec. 3 (norms).

13. The Cauchy–Schwarz inequality

The projection picture has one more thing to tell us, and it is a bound on the dot product itself.

Look again at the right triangle of §12.3. Its hypotenuse is $\mathbf{v}$, with length $\|\mathbf{v}\|$, and one of its legs is the shadow, with length $|\text{comp}_{\mathbf{u}}(\mathbf{v})|$. A leg of a right triangle is never longer than the hypotenuse — equivalently, and more memorably:

A shadow is never longer than the object casting it.

In symbols, $|\text{comp}_{\mathbf{u}}(\mathbf{v})| \leq \|\mathbf{v}\|$. Now substitute the definition of the component and clear the denominator:

$$\left| \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|} \right| \leq \|\mathbf{v}\| \quad \implies \quad |\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

That is a bound on how large a dot product can possibly be, in terms of nothing but the two magnitudes — and it is famous enough to carry two names.

Cauchy–Schwarz inequality. For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

There is also a one-line route to it from §11 that requires no picture at all. Since $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$ and $|\cos \theta| \leq 1$ for every angle,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\| \|\mathbf{v}\| |\cos \theta| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

This second route also settles when the bound is *tight*: equality needs $|\cos \theta| = 1$, hence $\theta = 0$ or $\theta = \pi$. So the dot product attains its largest possible size exactly when the two vectors are parallel or antiparallel — when, in the shadow picture, the triangle has collapsed and $\mathbf{v}$ lies flat along $\mathbf{u}$, casting a shadow as long as itself.

Example 13.1 (The bound in action). For $\mathbf{u} = [3, 1]$ and $\mathbf{v} = [1, 3]$ we found $\langle \mathbf{u}, \mathbf{v} \rangle = 6$ in §11, while $\|\mathbf{u}\| \|\mathbf{v}\| = \sqrt{10} \cdot \sqrt{10} = 10$. And $6 \leq 10$, with room to spare — the slack is exactly the factor $\cos \theta = 0.6$. For $\mathbf{u} = [1, 2]$ and $\mathbf{v} = [2, 4]$, which are parallel, we get $\langle \mathbf{u}, \mathbf{v} \rangle = 10$ and $\|\mathbf{u}\| \|\mathbf{v}\| = \sqrt{5} \cdot 2\sqrt{5} = 10$: equality, as predicted.

Remark (Why it matters more than it looks). Cauchy–Schwarz is what makes §11.4 legitimate. We defined $\cos(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle / (\|\mathbf{u}\| \|\mathbf{v}\|)$ and asserted that the output lands in $[-1, 1]$ — but rearrange the inequality and that assertion *is* Cauchy–Schwarz:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\| \iff -1 \leq \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1.$$

In $\mathbb{R}^2$ we could lean on the angle θ , which we can see. In higher dimensions, and in the abstract inner-product spaces of later chapters, there is no picture to look at and the angle is *defined* by that quotient — so something has to guarantee the quotient is a legal cosine before arccos may be applied to it. Cauchy–Schwarz is that guarantee, which is why it is stated as an inequality about inner products rather than as a fact about triangles.

Remark (A second dividend: the triangle inequality, again). In §7.2 we accepted $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ because a triangle cannot have one side longer than the other two combined — true, but geometric. It now follows algebraically. Expanding as in §11.2,

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2,$$

where the middle step is Cauchy–Schwarz. Taking square roots of both non-negative sides gives the triangle inequality. The geometric picture and the algebraic bound were the same statement all along.

14. Metrics, and the family of norms

Every norm we have written so far has been the Euclidean one, and we have been writing it $\|\mathbf{v}\|$ with no further comment. It is time to admit that this was a choice. Behind the norm sits a more primitive object — a notion of *distance* — and the Euclidean notion is only one of many.

14.1. Metrics

Definition 14.1 (Metric). Let A be a set. A **metric** on A is a function d that takes two elements of A and returns a non-negative real number — interpreted as the *distance* between them — subject to the following. For all $x, y, z \in A$:

M1. $d(x, y) = 0 \iff x = y$ (a point is at distance zero from itself, and from nothing else)

M2. $d(x, y) = d(y, x)$ (symmetry)

M3. $d(x, y) + d(y, z) \geq d(x, z)$ (triangle inequality)

M4. $d(x, y) \geq 0$ (non-negativity)

The elements of A are usually points of a coordinate space — for us $A = \mathbb{R}^2$, and the two elements are two vectors — but nothing in the definition requires that. Metrics are defined on sets of strings, of functions, of images, of DNA sequences. The four conditions are all that “distance” has to mean.

Remark. Axiom **M4** is listed for emphasis, but it is not independent: it follows from the other three. Put $z = x$ in **M3** to get $d(x, y) + d(y, x) \geq d(x, x)$, which is 0 by **M1**. Then **M2** turns the left side into $2d(x, y)$, so $2d(x, y) \geq 0$ and hence $d(x, y) \geq 0$. Worth knowing, because it means that to check something is a metric you only ever have to check three things.

14.2. The Euclidean distance is a metric

The distance we have been using all along is

$$d_2(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} = \|\mathbf{x} - \mathbf{y}\|,$$

the norm of the difference — which is exactly the between-the-heads reading of §7.3. Let us check all four axioms. Each one turns out to be a result we already have.

M1. Suppose $d_2(\mathbf{x}, \mathbf{y}) = 0$. Then $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 0$. A sum of two squares of real numbers is zero only if each square is zero, so $x_1 = y_1$ and $x_2 = y_2$ — that is, $\mathbf{x} = \mathbf{y}$ by the equality of §9. Conversely, if $\mathbf{x} = \mathbf{y}$ then both differences vanish and the square root is 0. ✓

M2. Since $\mathbf{x} - \mathbf{y} = -(\mathbf{y} - \mathbf{x})$, the scaling rule of §8 with $\alpha = -1$ gives

$$\|\mathbf{x} - \mathbf{y}\| = \|(-1)(\mathbf{y} - \mathbf{x})\| = |-1| \|\mathbf{y} - \mathbf{x}\| = \|\mathbf{y} - \mathbf{x}\|,$$

so $d_2(\mathbf{x}, \mathbf{y}) = d_2(\mathbf{y}, \mathbf{x})$. Geometrically: mirroring a displacement does not change its length. ✓

M3. This is §7.2 in disguise. Put $\mathbf{u} = \mathbf{x} - \mathbf{y}$ and $\mathbf{v} = \mathbf{y} - \mathbf{z}$, and notice that

$$\mathbf{u} + \mathbf{v} = (\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z}) = \mathbf{x} - \mathbf{z}.$$

The triangle inequality for norms says $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, which reads

$$\|\mathbf{x} - \mathbf{z}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\|, \quad \text{that is} \quad d_2(\mathbf{x}, \mathbf{z}) \leq d_2(\mathbf{x}, \mathbf{y}) + d_2(\mathbf{y}, \mathbf{z}).$$

✓

M4. A square root of a non-negative quantity is non-negative. ✓

Remark. Notice what **M3** explains retroactively. In §7.2 we called $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ the “triangle inequality” because of a picture of a triangle. We can now see the deeper reason for the name: it is the axiom that makes distance behave like distance, in *any* setting, with or without a picture.

14.3. A different geometry: the Manhattan metric

The Euclidean metric assigns to two points the length of the shortest straight line joining them. That is humanity’s most familiar notion of distance, and in flat coordinate space it is the right one. But “shortest path” depends on which paths are *available*, and that depends on the geometry of the space you are moving through.

So consider a plane carved into a grid of city blocks by streets and avenues, and suppose you must travel between two street corners. You are not allowed to cut through the buildings. Every

route is a sequence of moves along streets and avenues, and the Euclidean distance — the straight line between the corners — is not merely inconvenient but *wrong*: it would report the distance you could travel if you could pass through walls, which badly understates the trip.

What is the honest answer? Any legal route must cover the whole horizontal gap and the whole vertical gap, and no route need cover more. So the distance is the sum of the two gaps, and it does not matter in what order you take them.

Definition 14.2 (Manhattan metric). For $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ in $\mathbb{R}^2$, the **Manhattan** or **taxicab** metric is

$$d_1(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|.$$

The name is as literal as it sounds: it is the distance a taxi drives on a rectangular street grid.

Exercise (homework). Verify that d_1 satisfies **M1–M3**, and hence is a metric on $\mathbb{R}^2$.

Practice

P1. Compute $d_1(\mathbf{x}, \mathbf{y})$ for $\mathbf{x} = (1, 2)$ and $\mathbf{y} = (4, 6)$.

P2. Compute $d_1(\mathbf{x}, \mathbf{y})$ for $\mathbf{x} = (-2, 3)$ and $\mathbf{y} = (3, -1)$.

Answers.

$$\text{P1: } d_1 = |1 - 4| + |2 - 6| = 3 + 4 = 7, \quad \text{compare } d_2 = \sqrt{9 + 16} = 5.$$

$$\text{P2: } d_1 = |-2 - 3| + |3 - (-1)| = 5 + 4 = 9, \quad \text{compare } d_2 = \sqrt{25 + 16} = \sqrt{41} \approx 6.40.$$

In both cases the taxicab distance exceeds the Euclidean one, which is what you should expect: forbidding diagonal movement can only make the trip longer.

14.4. The Chebyshev metric

A third notion of distance keeps only the *largest* of the coordinate displacements and ignores the other entirely.

Definition 14.3 (Chebyshev metric). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$, the **Chebyshev** metric is

$$d_\infty(\mathbf{x}, \mathbf{y}) = \max\{|x_1 - y_1|, |x_2 - y_2|\}.$$

Remark (Why it is called the chess metric). Think of a king on a chessboard. A king moves one square per turn in any of eight directions, including diagonally — and a diagonal step advances it by one square *horizontally and vertically at the same time*. So the smaller of the two displacements comes for free: it is covered incidentally while closing the larger one. The number of moves a king needs to travel between two squares is therefore exactly the larger of the horizontal and vertical gaps — precisely d_∞ .

For example, from $(1, 1)$ to $(4, 3)$ the gaps are 3 and 2, so $d_\infty = 3$, and indeed three moves suffice: $(1, 1) \rightarrow (2, 2) \rightarrow (3, 3) \rightarrow (4, 3)$. No shorter route exists, because three columns must be crossed and a king crosses at most one per move.

14.5. Naming the norms

Each of these metrics has a norm attached to it, and the two are two views of the same measurement:

$$\|\mathbf{v}\| = d(\mathbf{v}, \mathbf{0}) \quad \text{and} \quad d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|.$$

The norm is the distance from the origin; the distance is the norm of the difference. So each of our three metrics names a norm, and since there are now several we can no longer write $\|\mathbf{v}\|$ and leave it at that. The standard fix is a **subscript**:

Norm	Notation	Formula
Manhattan / taxicab	$\ \mathbf{v}\ _1$	$ v_1 + v_2 $
Euclidean	$\ \mathbf{v}\ _2$	$\sqrt{ v_1 ^2 + v_2 ^2}$
Chebyshev	$\ \mathbf{v}\ _\infty$	$\max\{ v_1 , v_2 \}$

Remark (Our convention). Everywhere in §1–§13, and everywhere later unless a subscript says otherwise, a bare $\|\mathbf{v}\|$ means the Euclidean norm $\|\mathbf{v}\|_2$. The Euclidean norm keeps the privilege of needing no subscript.

Which leaves an obvious question: why 1, 2 and ∞ ? The numbers are not arbitrary labels. They are parameter values.

14.6. The p -norm

Definition 14.4 (p -norm). For a real number $p \geq 1$ and $\mathbf{v} = [v_1, v_2] \in \mathbb{R}^2$, the p -**norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\|_p = (|v_1|^p + |v_2|^p)^{1/p}.$$

Two of our three norms are already on this list. Setting $p = 1$,

$$\|\mathbf{v}\|_1 = (|v_1|^1 + |v_2|^1)^{1/1} = |v_1| + |v_2|,$$

which is the Manhattan norm. Setting $p = 2$,

$$\|\mathbf{v}\|_2 = (|v_1|^2 + |v_2|^2)^{1/2} = \sqrt{v_1^2 + v_2^2},$$

which is the Euclidean norm — the absolute values are harmless here, since squaring destroys signs anyway. So the subscripts were never names; they were *the value of p* . This is why the Manhattan and Euclidean norms are properly called the **1-norm** and the **2-norm**.

Remark (Why $p \geq 1$). The restriction is not decoration — the formula stops being a norm below 1. Take $p = \frac{1}{2}$, $\mathbf{u} = [1, 0]$ and $\mathbf{v} = [0, 1]$. Then $\|\mathbf{u}\|_{1/2} = 1^2 = 1$ and $\|\mathbf{v}\|_{1/2} = 1$, while $\mathbf{u} + \mathbf{v} = [1, 1]$ gives

$$\|\mathbf{u} + \mathbf{v}\|_{1/2} = (1 + 1)^2 = 4.$$

But $4 > 1 + 1$, so the triangle inequality fails. Whatever that formula measures, it is not a length.

14.7. What happens as $p \rightarrow \infty$

That accounts for 1 and 2. But the Chebyshev norm was defined by a *maximum*, which looks nothing like the p -norm formula — and its subscript is not even a number. Both puzzles are resolved by the same computation.

Claim. For every $\mathbf{v} \in \mathbb{R}^2$,

$$\lim_{p \rightarrow \infty} \|\mathbf{v}\|_p = \max\{|v_1|, |v_2|\}.$$

Proof. If $\mathbf{v} = \mathbf{0}$ both sides are 0, so assume $\mathbf{v} \neq \mathbf{0}$ and write

$$M = \max\{|v_1|, |v_2|\} > 0.$$

The idea is to trap $\|\mathbf{v}\|_p$ between two quantities that both approach M , and then apply the squeeze theorem from MATH 141.

A lower bound. One of the two coordinates realises the maximum, so one of the terms $|v_1|^p, |v_2|^p$ equals M^p exactly. The other term is non-negative. Hence

$$|v_1|^p + |v_2|^p \geq M^p,$$

and raising both sides to the power $1/p$ — which preserves the inequality, since $t \mapsto t^{1/p}$ is increasing on $[0, \infty)$ —

$$\|\mathbf{v}\|_p \geq (M^p)^{1/p} = M.$$

An upper bound. Every coordinate is at most M in absolute value, so each term is at most M^p , and there are two of them:

$$|v_1|^p + |v_2|^p \leq 2M^p.$$

Again taking p -th roots,

$$\|\mathbf{v}\|_p \leq (2M^p)^{1/p} = 2^{1/p} M.$$

Squeezing. Combining the two bounds, for every $p \geq 1$,

$$M \leq \|\mathbf{v}\|_p \leq 2^{1/p} M.$$

Now let $p \rightarrow \infty$. The left-hand bound does not depend on p at all, so it stays at M . On the right, $1/p \rightarrow 0$ and $t \mapsto 2^t$ is continuous, so $2^{1/p} \rightarrow 2^0 = 1$ and the whole bound tends to M . Both outer quantities converge to the same limit M , so by the squeeze theorem the quantity trapped between them does too:

$$\lim_{p \rightarrow \infty} \|\mathbf{v}\|_p = M = \max\{|v_1|, |v_2|\}. \quad \blacksquare$$

Both puzzles dissolve at once. The Chebyshev norm is not a separate construction bolted on beside the others — it is the *limiting case* of the p -norm family, which is exactly why its subscript is ∞ :

$$\|\mathbf{v}\|_\infty := \lim_{p \rightarrow \infty} \|\mathbf{v}\|_p = \max\{|v_1|, |v_2|\}.$$

And the maximum was inevitable. As p grows, raising to the p -th power exaggerates the gap between a larger and a smaller number without limit, so the larger coordinate comes to dominate the sum completely and the smaller one becomes negligible. Taking $p \rightarrow \infty$ makes that domination total.

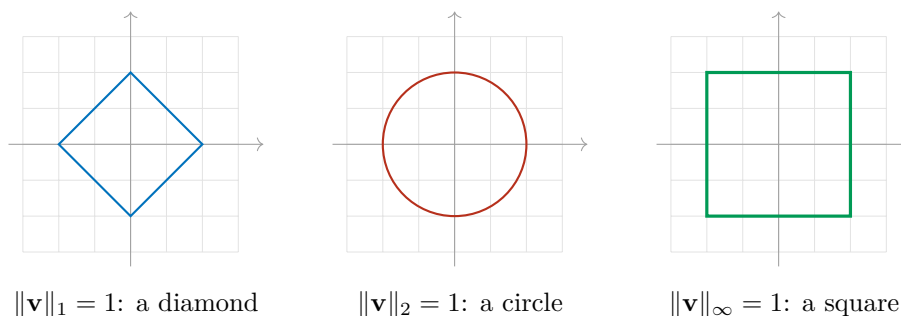
Remark. Nothing in the argument used the fact that there are exactly two coordinates. In $\mathbb{R}^n$ the upper bound becomes $n^{1/p}M$, and $n^{1/p} \rightarrow 1$ just as $2^{1/p}$ did — the constant is different, the limit is not. All that matters is that the number of coordinates is finite.

14.8. Unit balls: the shape of “length one”

In §12.1 we found that the vectors of unit length trace a circle of radius 1, and flagged that this was an artefact of the Euclidean norm. Now we can see what the other norms give. In each case we are drawing the same set — the vectors whose length is 1 — under a different definition of length.

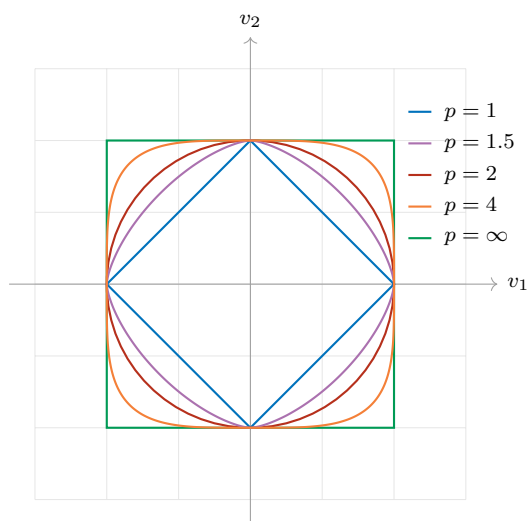
- $\|\mathbf{v}\|_1 = 1$ means $|v_1| + |v_2| = 1$. In the first quadrant that is the line $v_1 + v_2 = 1$ joining $(1, 0)$ to $(0, 1)$; reflecting into the other three quadrants gives a **diamond** with vertices at $(\pm 1, 0)$ and $(0, \pm 1)$.
- $\|\mathbf{v}\|_2 = 1$ means $v_1^2 + v_2^2 = 1$: the **circle** of radius 1.
- $\|\mathbf{v}\|_\infty = 1$ means the larger of $|v_1|, |v_2|$ equals 1 while the other is at most 1: the boundary of the **square** with corners $(\pm 1, \pm 1)$.

All three are centred at the origin. The set is called the *unit ball* of the norm, and its shape is the norm’s signature.



Varying p continuously

The three pictures are not three unrelated shapes. The p -norm is continuous in p , so as p increases from 1 the unit ball deforms smoothly: the diamond’s flat edges bow outward, passing exactly through the circle at $p = 2$, and continue bulging until they flatten against the square in the limit $p \rightarrow \infty$.



Reading the picture from the inside out also reveals an ordering. The diamond sits inside the circle, which sits inside the square, and the balls only ever grow as p grows. Since a *larger* unit ball means it takes a bigger vector to reach length 1, the norms themselves run the other way:

$$\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1.$$

Check it on $\mathbf{v} = [1, 1]$: the three norms are 1, $\sqrt{2} \approx 1.414$, and 2. The same vector has three different lengths, and which one is “correct” is a question about the problem you are solving, not about the vector.

Looking ahead. We have now met four metrics and a whole family of norms, but we have not yet said what a *norm* is in the way that §14.1 said what a metric is — the $p \geq 1$ restriction, for instance, was justified by one counterexample rather than by a definition. The next section gives the axioms a norm must satisfy and checks them against the family we have built.

15. What makes a norm a norm

Section 13 built an entire family of norms without ever saying what the word means. A **norm** is a function that takes a vector and returns a non-negative real number,

$$\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R},$$

subject to three conditions. Every member of the p -norm family satisfies them, for every p in $[1, \infty)$ — and so does the limiting case $p = \infty$.

15.1. The three properties

For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ and all $\alpha \in \mathbb{R}$:

N1. Positive definiteness.

$$\|\mathbf{v}\| \geq 0, \quad \text{and} \quad \|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}.$$

N2. Absolute homogeneity.

$$\|\alpha\mathbf{v}\| = |\alpha| \|\mathbf{v}\|.$$

N3. Triangle inequality (also called *subadditivity*).

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

None of these should be new. **N2** is the scaling rule we established in §8 — the size $|\alpha|$ sets the length and the sign is discarded, which is exactly why the absolute value appears. **N3** is §7.2, first justified by a triangle and later derived from Cauchy–Schwarz in §13. What §15 adds is the recognition that these are not three assorted facts about the Euclidean norm; they are the *definition* of length, and any function satisfying them earns the name.

Remark (Which parts are doing real work). Exactly as with the metric axioms of §14.1, the list is not minimal. Homogeneity with $\alpha = 0$ gives $\|\mathbf{0}\| = |0|\|\mathbf{v}\| = 0$, so half of **N1** is free. And non-negativity follows from the other two:

$$0 = \|\mathbf{0}\| = \|\mathbf{v} + (-\mathbf{v})\| \leq \|\mathbf{v}\| + \|-\mathbf{v}\| = \|\mathbf{v}\| + |-1|\|\mathbf{v}\| = 2\|\mathbf{v}\|,$$

so $\|\mathbf{v}\| \geq 0$. What cannot be derived, and is therefore the real content of **N1**, is the converse direction: $\|\mathbf{v}\| = 0$ forces $\mathbf{v} = \mathbf{0}$. Drop it and the function sending *every* vector to 0 would qualify as a norm — it satisfies **N2** and **N3** perfectly. That is why the property is called *positive definiteness* rather than merely non-negativity: it is the clause that stops a norm from collapsing.

15.2. The family passes

Two of the three are quick to confirm for every p at once.

N1. $\|\mathbf{v}\|_p = 0$ forces $|v_1|^p + |v_2|^p = 0$. That is a sum of two non-negative numbers, so both vanish, so $v_1 = v_2 = 0$. ✓

N2. Pull the scalar out one coordinate at a time:

$$\|\alpha \mathbf{v}\|_p = (|\alpha v_1|^p + |\alpha v_2|^p)^{1/p} = (|\alpha|^p (|v_1|^p + |v_2|^p))^{1/p} = |\alpha| (|v_1|^p + |v_2|^p)^{1/p} = |\alpha| \|\mathbf{v}\|_p.$$

✓

N3 is the hard one, and it is precisely where the restriction $p \geq 1$ earns its keep — recall the $p = \frac{1}{2}$ counterexample of §14.6. The two ends of the family are easy. For $p = 1$, apply the scalar inequality $|a + b| \leq |a| + |b|$ in each coordinate:

$$\|\mathbf{u} + \mathbf{v}\|_1 = |u_1 + v_1| + |u_2 + v_2| \leq |u_1| + |v_1| + |u_2| + |v_2| = \|\mathbf{u}\|_1 + \|\mathbf{v}\|_1.$$

For $p = \infty$, the same scalar inequality inside a maximum:

$$\|\mathbf{u} + \mathbf{v}\|_\infty = \max_i |u_i + v_i| \leq \max_i (|u_i| + |v_i|) \leq \max_i |u_i| + \max_i |v_i| = \|\mathbf{u}\|_\infty + \|\mathbf{v}\|_\infty.$$

For p strictly between, the inequality is a genuine theorem with a name — **Minkowski's inequality** — whose proof needs machinery beyond this course. We will use it, and record that it holds for every $p \geq 1$ and fails for every $p < 1$.

15.3. Cauchy–Schwarz, generalised

In §13 we proved $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$, with both norms Euclidean. Now that other norms exist, the natural question is what replaces it.

Remark (The obvious guess is false). One might hope for $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|_p \|\mathbf{v}\|_p$ with the *same* p on both factors. Test it at $p = \infty$ with $\mathbf{u} = \mathbf{v} = [1, 1]$: then $\langle \mathbf{u}, \mathbf{v} \rangle = 2$, while $\|\mathbf{u}\|_\infty \|\mathbf{v}\|_\infty = 1 \cdot 1 = 1$. The bound would claim $2 \leq 1$. So the two exponents cannot be equal in general.

The correct statement pairs each exponent with a partner.

Hölder's inequality. Let $p, q \geq 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|_p \|\mathbf{v}\|_q.$$

Such a p and q are called **conjugate exponents**. Three pairings matter in practice:

$$(p, q) = (2, 2), \quad (p, q) = (1, \infty), \quad (p, q) = (\infty, 1),$$

where the convention $1/\infty = 0$ makes the middle two legitimate. And notice the first: $\frac{1}{2} + \frac{1}{2} = 1$, so $p = q = 2$ is conjugate to itself, and Hölder reads

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|_2 \|\mathbf{v}\|_2,$$

which is Cauchy–Schwarz exactly. So Cauchy–Schwarz was never a standalone result; it is the self-conjugate case of a family, and the reason it looks symmetric is that 2 is the one exponent that partners with itself. Re-running the failed example above with the correct pairing $(\infty, 1)$ gives $\|\mathbf{u}\|_\infty \|\mathbf{v}\|_1 = 1 \cdot 2 = 2$, and $2 \leq 2$ holds — with equality.

15.4. Norms beyond $\mathbb{R}^2$: quaternions

Nothing in **N1–N3** mentioned the dimension, or even that the objects were arrows in a plane. Anything with a sensible notion of addition and scaling can carry a norm. The quaternions of the complex numbers lecture are a convenient example: a quaternion

$$q = a + bi + cj + dk$$

carries four real numbers, so it behaves like a vector in $\mathbb{R}^4$, and its modulus is the Euclidean norm of that 4-vector:

$$|q| = \sqrt{a^2 + b^2 + c^2 + d^2} = \|[a, b, c, d]\|_2.$$

All three properties come for free, because it *is* a Euclidean norm: a sum of four squares vanishes only when every term does (**N1**), a real scalar factors out in absolute value (**N2**), and **N3** is the triangle inequality in $\mathbb{R}^4$.

Example 15.1. For $q = 1 + i + j + k$, $|q| = \sqrt{1+1+1+1} = 2$. For $q = 3 + 4i$ — a complex number, viewed as a quaternion with $c = d = 0$ — $|q| = \sqrt{9+16} = 5$, agreeing with the modulus we would have computed in $\mathbb{C}$.

Remark (One property vectors do not have). The quaternion norm is **multiplicative**: $|q_1 q_2| = |q_1| |q_2|$. Check it on the product computed in the complex numbers lecture, $q = 1 + i$ and $p = 1 + j$, whose product was $qp = 1 + i + j + k$:

$$|q| |p| = \sqrt{2} \cdot \sqrt{2} = 2 = |1 + i + j + k| = |qp|. \quad \checkmark$$

No such statement is available for vectors in $\mathbb{R}^2$, for the simple reason that there is nothing to put on the left: two vectors have no product that returns a vector. Multiplicativity is a property of the extra algebraic structure, not of the norm.

15.5. Why this matters

Norms are among the most heavily used operations in computational science, and they will recur throughout this course — rarely as an object of study in their own right, and almost always as the thing being measured, minimised, or bounded. A norm is how we say that an approximation is close to an answer, that an algorithm has converged, that an error is small enough to stop, or that one matrix is nearly another.

Looking ahead. Three appearances worth anticipating. Least squares is stated as $\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2$ — a norm being minimised, and the choice of $p = 2$ is exactly what makes the problem solvable by the geometry of §12.6. Every iterative method needs a stopping rule of the form “halt when $\|\text{residual}\|$ drops below a tolerance,” so the norm sets the meaning of *done*. And the notion of a norm extends from vectors to matrices, giving the condition number that measures how badly a linear system amplifies error.

Looking ahead. A last remark on the unit-ball pictures of §14.8, which are less decorative than they look. The 1-norm’s diamond has its corners *on the axes*, where one coordinate is zero; the 2-norm’s circle has no corners at all. When a norm is minimised subject to a constraint, solutions tend to land at corners — so minimising with $\|\cdot\|_1$ tends to produce answers with zero entries, while $\|\cdot\|_2$ does not. That single geometric difference is the reason ℓ_1 penalties are used to force sparse solutions in modern statistics and machine learning. The shape of the ball is the whole explanation.